

Homework (Carolina Reaper)

Q1. Solve for x : $2^x = 32$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q2. A musician is creating a rhythm pattern with a frequency of 3^x beats per minute. If the frequency is 27 beats per minute, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. Solve for y : $4^y = 64$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q4. An artist is creating a canvas with a size of 2^x inches by 3^x inches. If the area of the canvas is 48 square inches, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$

(E) $x = 5$

Q5. Solve for z : $5^z = 125$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q6. A sound wave has a frequency of 2^x Hz. If the frequency is 16 Hz, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q7. Solve for w : $3^w = 81$

- (A) $w = 1$
- (B) $w = 2$
- (C) $w = 3$
- (D) $w = 4$
- (E) $w = 5$

Q8. A painter is mixing colors with a ratio of $2^x : 3^x$. If the ratio is 4 : 9, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Open Ended Questions (FRQ)

Q9. Solve for x : $x^2 + 2^x = 12$. Show all work and explain your reasoning.

Q10. A music producer is creating a remix with a tempo of 3^x beats per minute. If the tempo is increasing by a factor of 2 every 2 minutes, and the initial tempo is 9 beats per minute, find the value of x after 4 minutes. Show all work and explain your reasoning.

Chef's Tips

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- Tip 1: Encourage students to use properties of exponents to simplify equations
- Tip 2: Remind students to check their work and verify their solutions
- Tip 3: Emphasize the importance of explaining their reasoning and showing all work